UNIVERSITY OF LONDON

BSc EXAMINATION 2024

For Internal Students of
Royal Holloway

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IY3609: DIGITAL FORENSICS

IY3609R: DIGITAL FORENSICS – PAPER FOR FIRST SITS/RESIT CANDIDATES

Time Allowed: **TWO** hours

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Attempt **three** questions.
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Question 1 Legal and Procedural Matters

(a) (i) How do the standards of evidence differ in criminal and civil cases? [2]

(ii) What is referred to as a "parallel construction", which is a practice that is arguably allowed in the UK under the Investigatory Powers Act 2016? Define the term and provide a short example. [2]

(iii) Describe two tests to distinguish between technical evidence and expert evidence (following an Australian decision in the case *The Queen v. Bonython (1984)*). [2]

(b) Kevin is a police officer who is in charge of searching a suspect's apartment. When he enters the apartment, he finds the suspect's desktop computer powered on and unlocked. He immediately cuts the computer's power, removes the hard drive, and places it in his backpack. Back in his office, using a SATA-to-USB adapter, he connects the hard drive to a laptop running Arch Linux and starts looking for suspicious content using the Linux command *grep*. When he finds a text file containing the word "bombing", he copies this file to his laptop using the Linux command *cp*.

Point out five mistakes that Kevin made that put the (continuity of) evidence at risk. Make suggestions how he should have proceeded instead (based on principles outlined by, e.g., the ACPO Guidelines). [10]

(c) Serena is suspected to have participated in a spectacular theft of military firearms in preparation for a terrorist attack. However, she claims to have slept peacefully the entire night on the date in question. The prosecution wants to use activity and location data collected on her smartwatch as proof of her illicit nightly activities. The particular smartwatch model stores this data on the device itself and on the connected smartphone, which in turn synchronizes it with cloud servers located in the US.

Discuss three applicable legal instruments that could be used to obtain Serena's data from the cloud servers. [9]

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Question 2 Storage Forensics 1

(a) (i) How can a forensic analyst identify which file systems are present on a disk by inspecting the MBR? [2]

(ii) Describe two ways to establish that a volume holds a FAT file system if there is no MBR or GPT present. [2]

(iii) Give two examples of relevant information a forensic analyst can learn by reading out SMART diagnostic information of a drive. [2]

(b) Sam is analysing a drive that contains a damaged FAT32 file system that uses a cluster size of 4KB. He is trying to manually reconstruct a 19KB file "evidence.txt", for which he found the corresponding directory entry.

(i) The directory entry contains the following information:

- `First cluster (MSB): 0x0000`
- `First cluster (LSB): 0x000C`

Calculate the number of the first cluster in decimal representation. Describe and document all involved intermediate steps. [4]

(ii) Based on the calculation above and the file allocation table given below, is it possible to identify all clusters of the file "evidence.txt"? If so, list *all* clusters of the file. Otherwise, argue why not and discuss an alternative approach to recover the file. [5]

0 1	Volume Info
2	10
3	8
4	6
5	EOC
6	*Unreadable*
7	9
8	EOC
9	3
10	EOC
11	14
12	7
13	*Unreadable*
14	*Free*
...	...

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(c) Charlotte is working in a sophisticated forensic lab and is tasked with recovering deleted data from an SSD. The SSD has a single partition that is formatted with FAT32.

Advise how Charlotte should approach her task. Specifically, discuss issues related to creating a forensic image of the SSD in the likely presence of garbage collection mechanisms. Then, suggest a suitable procedure to mitigate these issues. [10]

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Question 3 Storage Forensics 2

(a) (i) Briefly describe what is referred to as "cryptographic erasure". Why is this data erasure method so effective to counter forensic analyses? [2]

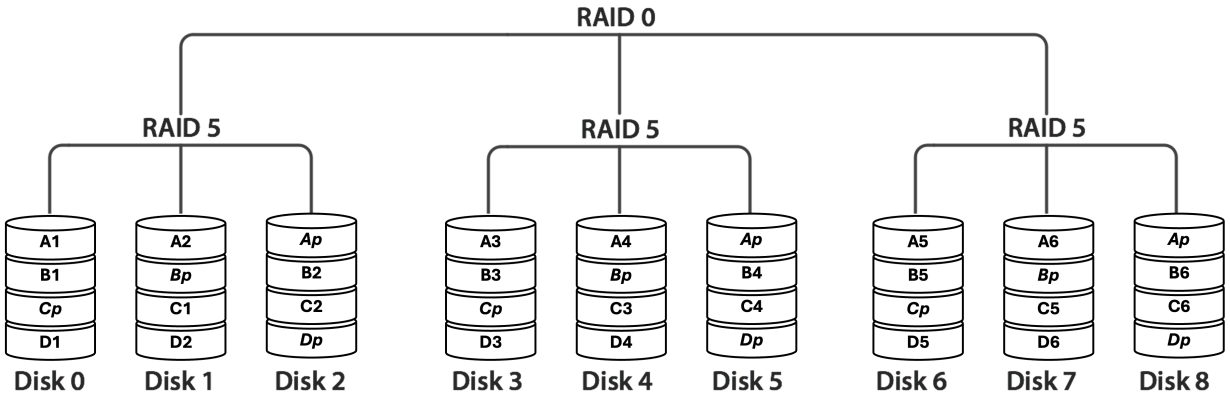
(ii) Which type of forensic analysis does file-level encryption prevent? Why might volume-level encryption be used in addition to file-level encryption to counter forensic analyses? [2]

(iii) For a damaged 3-disk storage array configured with RAID level 1, why is analysing 1 out of 3 disks sufficient to reconstruct all data? [2]

(b) Sarah works in a forensic lab and is tasked with recovering data from a seized SSD. She is told that the owner of the drive during interrogation has bragged about his high security standards. In fact, he did protect the relevant data with three layers of encryption: drive- (TCG Opal 2.0), volume- (Microsoft BitLocker), and file-level encryption (Windows EFS).

For each level of encryption, propose one approach how Sarah could try to circumvent it to ultimately obtain the owner's data in the clear. For each proposed approach, state two assumptions for it to work out. [9]

(c) Paul is a forensic expert assisting a small company in disaster recovery. The data centre in the basement of the main building was badly hit by a fire. Their only storage server with 8 disks was operating in a RAID 50 configuration as shown below.



RAID 50 configuration (source: wikipedia.org)

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(i) Assume disks 0, 5, and 7 were completely destroyed by the fire. Is it still possible to fully recover all data from the remaining disks? If yes, provide a detailed reasoning why. If not, argue whether it's possible to at least partially recover data. [7]

(ii) How does this nested configuration compare to regular RAID 5 in terms of fault tolerance? [3]

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Question 4 Host Forensics

(a) (i) Name and briefly discuss two problems of live imaging techniques. [2]

(ii) Briefly describe two counter-forensic strategies to prevent or fool live acquisition attempts. [2]

(iii) Which information does the `HKEY_LOCAL_MACHINE` section of the Windows Registry store and why might this data be of interest for forensic analyses? [2]

(b) Paola is a forensic specialist assisting in a police investigation. She plans to create a memory image of a live system running Microsoft Windows using an external acquisition technique. Specifically, she attempts to obtain memory access by exploiting DMA via Thunderbolt.

(i) Describe how Paola this way can create a memory image of the entire system. Provide a brief discussion how DMA works and how it can be exploited for this purpose. [4]

(ii) The system is equipped with an IOMMU. How does this impact her chances for the memory acquisition to be successful? Focus your discussion on the functioning of an IOMMU and the key requirement for it to be effective. [5]

(c) Emmanuel is a security investigator in a large company. He is working on a case where an employee of the company is suspected to have abused access to a powerful virtual machine for mining cryptocurrencies at night.

(i) Assume the virtual machine was very recently shut down but not deleted from the virtualization server. Discuss two ways how Emmanuel can try to find evidence for the employee's crypto mining activities. [6]

(ii) Assume the virtual machine is still running. Discuss one way (different from the ones discussed above) how Emmanuel can try to find evidence for the employee's crypto mining activities. [4]

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Question 5 Mobile Forensics

(a) (i) Give two examples of information that may be retrieved by analysing a SIM card. [2]

(ii) What is referred to as "JTAG analysis"? Briefly outline how this method works in the context of mobile forensics. [2]

(iii) Why is a "chip-off" analysis likely to not yield useful results for recent smartphone generations? [2]

(b) Eva is working in a forensic lab. She is tasked with extracting the list of last received calls from an Android phone that is locked but not encrypted. The phone has fingerprint scanning and facial recognition enabled, and supports PIN-based authentication as a fallback.

(i) Suggest three ways that Eva can try to unlock the phone. Briefly describe each approach and state one key requirement for it to be successful. [6]

(ii) Assume Eva was not able to bypass the lock screen. Describe an alternative way she can get access to data on the phone. Once successful, where can she find the desired information about received calls? [4]

(c) Harry is a forensic expert who is supposed to investigate whether there is evidence that a suspect in a criminal case likely was present at a specific location at a specific time. For this, Harry is supposed to analyse GPS data recorded on the suspect's mobile phone through a fitness tracking application. During his investigation, Harry notices that the Android phone is infected by multiple malicious applications. In fact, one malware turns out to be a rootkit.

(i) Assume that Harry can extract the requested GPS data from the Android phone. Could it be possible that the malware has manipulated the GPS data? Provide a reasoned discussion, considering also the special case of a rootkit being present. [5]

(ii) Which counter-forensic techniques could hinder his rootkit analysis? Describe one example. [4]

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